McMaster University

Introduction to 3RA3

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Lecture #01
CS/SE 3RA3 - Fall 2025

McSCert

cc BY NC SA

Sébastien Mosser

Snowboarding *since 1995.* ***Composing*** *things since 2007.*

McSCert

- **22-...:** Associate Professor, McMaster Univ.
- **19-21:** "*Professeur*", UQAM
- **12-18:** "*Maître de Conférences*", Univ. Côte d'Azur
- **11-12:** Research Scientist, SINTEF
- **10-11:** Postdoc, Inria Lille Nord-Europe
- **07-10:** PhD student, Université de Nice

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3RA3 in a Nutshell

Requirements?

2

3

Ok, this is 100% useless

CS/RE 3RA3 In a Nutshell

Quick overview and logistic information

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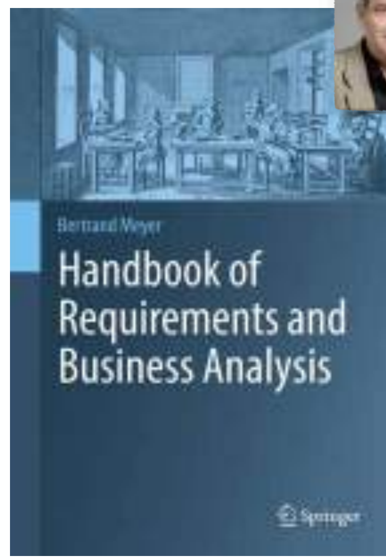
It takes a village to build a course

Dr. Mireille Blay-Fornarino

Dr. Richard Paige

Dr. Jean-Michel Bruel

And discussions with Dr. Daniel Amyot (uOttawa) and Dr. Gunter Mussbacher (McGill)



Our Amazing Team!

Sophia

Alina

Sathurshan

Kalvin

Nirmal

Mina

Yours Truly

Why am I teaching this course?

- **Dr. Paige** is now **Department Chair** at CAS
  - *Off-the-record: And no one else wanted to teach it ("It's hard", "Students hate it", …)*
- **Requirements-driven industrial projects**
  - Collaboration with companies like **TELUS**, **OSEAN**, **Visteon**, **EIT Digital**
  - ***Software***, ***embedded*** systems, ***mental health*** and ***ageing population***
- **10+ years of industrial consulting**
  - Software Architecture, ***Agile Requirements Elicitation***, Spec. ***Recruitment***
  - Founder of the **Agile Playground** in France (Polytech'Nice/Goood)

Evaluations

ID	Assignment	Kind	Weight	Start date	Delivery date
P	ACME Project	*Team*	40%	02/09/25	04/12/25
M	Midterm (MCQ)	*Individual*	15%	20/10/2025, 16:30	
F	Final exam	*Individual*	45%	*To be announced by registrar*	

- **It's your first time** writing requirements! **You'll make mistakes**. A lot.
  - The project is organized to **give you chances to fix your mistakes**
    - Two "**Window of Opportunity**" to get **feedback** and "**estimated grade**".
- **Midterm** evaluates **fundamental knowledge** AND **analysis skills** (MCQ)
  - *MSAF-ing the midterm carries on the weight to the final*
- The **exam** is **comprehensive** (work on your project ⇒ exam passed in a nutshell)

You will make

A lot of mistakes

Aka “Failure is Inevitable”

Wisdom from a Jedi Requirements Master

*You should start by **reading the scenario in detail**. It is **intentionally brief**, **intentionally slightly ambiguous**, and is meant to give you the **opportunity to try some of the techniques and methods** you've learned in the lectures and tutorials.*

*What **matters most** in doing the assignment is your **explanation** and **justification**, not whether you produce “**the right answer**”. As with most requirements problems, **there is more than one good answer.** What we are looking for is **well-justified answers** that at the very least **correctly apply the techniques** (modelling languages, tools, methods) covered in lectures.*

— Dr. Richard Paige

The "Case Study" approach

- You'll work on the **ACME, Connect!** Case study
  - **Helping newcomers** join University life.
  - You'll go through **three (3) cycles of requirements** elicitation (no code involved)
    - And yes, **there's enough content to keep you busy for 13 weeks** (teams of 3)
- **Req. Eng. is new for you**: we're here to **help** you and **support** you
  - Your customer is available for "**Ask Me Anything**" sessions on Teams (1/week)
  - We have designed for you an example project: **ATCO Eats**
    - We'll use this project to illustrate **good**, **bad**, and **terrible** requirements

Requirements?

What the hell are we talking about?

2

People do not buy products

- They buy **solutions** to their **problems**.
- How do we **characterize such solutions**?
  - That's **precisely what RE is** all about
- What we'll cover:
  - How **successful RE** makes a difference in practice.
  - How requirements are **expressed**? **Managed**?
  - **Role of security** in RE & product development

*It's an **intro course**. We're targeting "**Breadth of knowledge**"*

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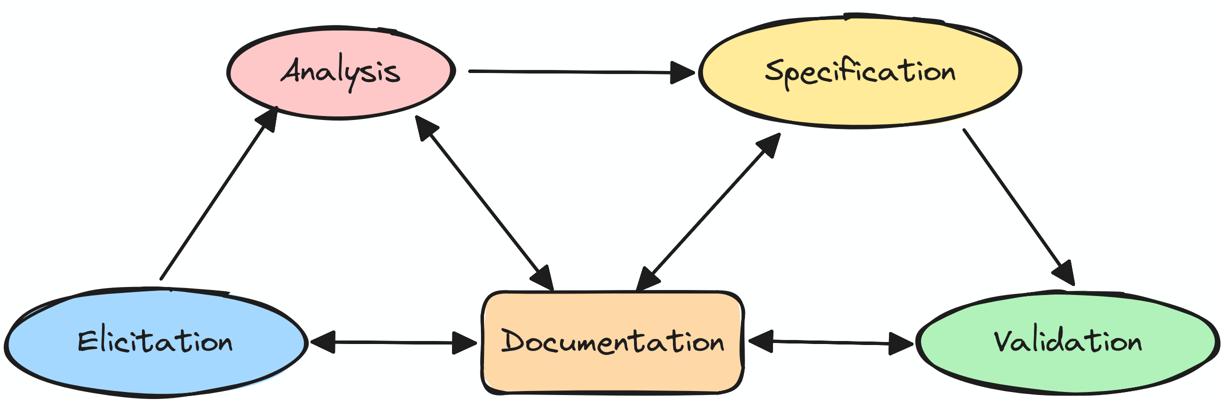
The Requirements Roadmap



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Why should I care?

- Quote: "**No one use Requirement Engineering in real life**"
  - Quote bis: "…***and it's usually a catastrophe***"
- Between **40-60% of software project fail**
  - Or **objectively struggle** because of poor requirements
- Around **50% of projects exceed their budget by 175% or more** because of poor requirements
  - **80% of software rework** can be traced to poor requirements
  - **50% of software defects** can be traced to poor requirements

Examples of Requirements

- **Things a product must do**
  - *The product shall produce an amended road de-icing schedule when a change to a truck status means that previously scheduled work cannot be carried out as planned.*
- **Qualities a product must have** (e.g., fast, secure, usable)
  - *The product shall use company colours, standard company logos and standard company typeface*
- **Constraints a product must obey** (standards, laws)
  - *The product will run on the department existing Windows Machine.*

Good requirements in a nutshell

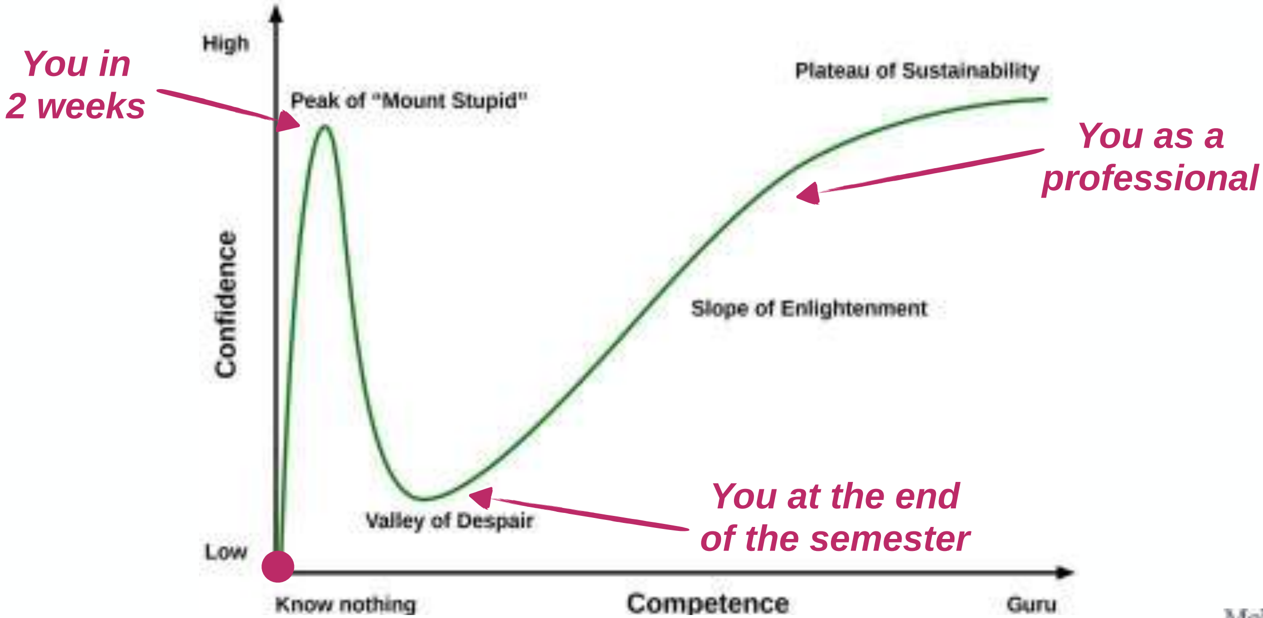
- A **good requirement** states something that is
  - **necessary**,
  - **verifiable**,
  - And **attainable**.
- "*The system shall instantly generate a giant Gummy Bear every time the player starts Baldur's Gate 3*": **Absolutely necessary**, **easily verifiable**. Attainable?
- What would be a **better requirement**?

Bad Requirements Engineering

A bad requirement is not *necessary*, *verifiable* or *attainable*

(And RE is not about saying "Yes" to every customer request)

“3RA3 could have been a youtube video”



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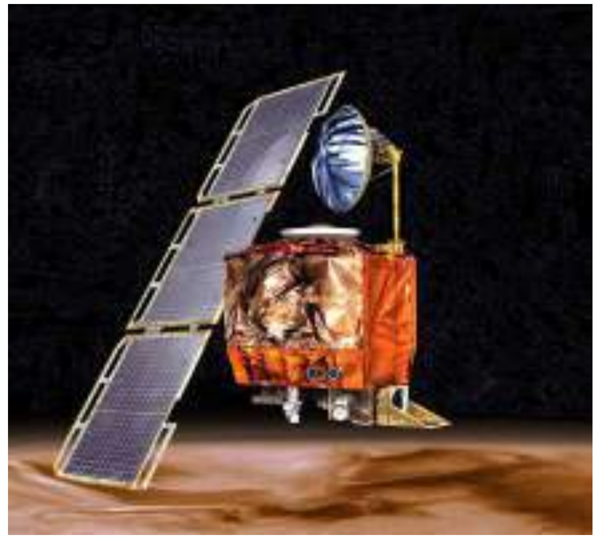
Requirements are useless

I mean, “the code is the truth”, right?

Space Exploration Fiascos

- **Ariane V - Flight v88** (1st flight)
  - Went kaboom after 37s
  - Estimated cost: **$370M USD**
    - Inflation + exchange: **$1B CAD**
  - **Bad code reuse, dead code, overflow,…**
- **Mars Climate Orbiter** (lost after 11 months)
  - **$110M USD** to build and deploy
  - **Unit conversion error** ($lb.s^2 \neq N.s^2$)



I'm not going to space, I want to do AI!

- **Tesla auto-pilot** anti-collision not able to identify black pedestrians at night
- **Amazon HR** recruitment assistant amplifying Ivy League recruitment biases
- **Google Photo** recognizing black people as gorillas or chimpanzees
- **Contact tracing** application during COVID-19
- **Microsoft navigation** system patented an "avoid ghetto" mode for pedestrian walks

No space, no AI. Automotive?

(No one was hurt, even if the video is impressive)

- **Conflicting Requirements**
  - **Pedestrian Anti-collision**
    - *Detecting a pedestrian implies stopping the vehicle*
  - **Traffic collision avoidance**
    - *Accelerating strongly bypasses other subsystems to support escaping maneuvers.*

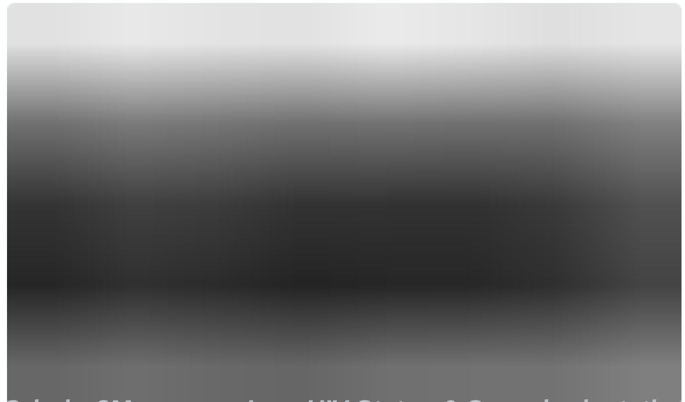
Well, … Automotive is a no. Video-games!

- Diablo 4 players praised Path of Exile's design philosophy, stating that it represented an ideal version of what Diablo should be, especially in terms of skill usage.
- Path of Exile developers criticized the use of cooldowns as a solution to encourage players to use different skills, describing it as "rotational behavior" that lacks strategic gameplay.
- The developers also condemned the reliance on skills that consistently gain resources to use other more powerful skills, which is basically how the Sorcerer class works in Diablo 4.

- **Blizzard** (Diablo) approach:
  - "We are experts, **we know better**"
- **Grinding Gears Game** (PoE) approach:
  - Expertise coupled to an **alpha/closed-beta/open-beta cycle** (3 years)
- The **Hades** game on the **Switch** used a similar cycle

I don’t care, I want to be a mobile developer!

Waze Drive Now: 151M users — Home Location



Grindr: 2M users — Loc, HIV Status & Sexual orientation

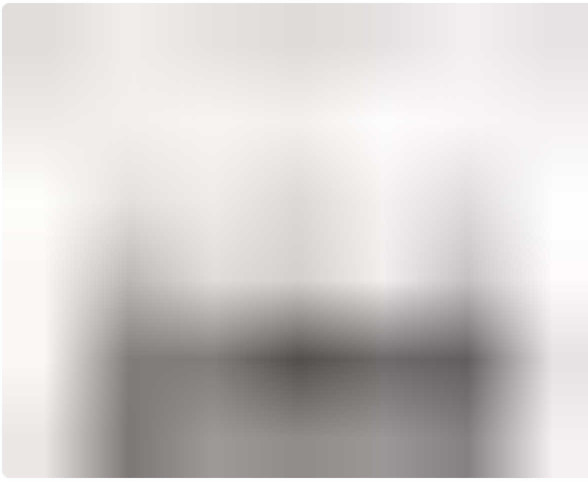
***Leaking personal information**, **sold to** external **companies**.*

No mobile then. Plain old server-side.

77M of PSN accounts stolen

4.2M of personal banking info stolen

You're exaggerating. It can't be that bad.



Cost: €7M. 120k soldiers unpaid for years.

Cost: $5B. 100k+ civil servants unpaid.

Payroll regulations are so complex that **software cannot compute proper payslips**.

No one care about RE, everyone is "vibe coding"

Conclusions

TL;DR: Takeaway Messages

A whole new world!

Welcome to the Human dimension of Soft. Eng.
*Spoiler alert: Humans s*ck.*

Will AI take your job?

If your only skill is to write code?

Will AI take your job?

If you understand requirements engineering?

Takeaway Message

- **One cannot escape requirements engineering**
  - **Requirements are everywhere**
- **Fixing a problem** at the requirement level
  - It is **easier** and **costs less**
  - Requires **eng. methods** to be identified
- **Requirements Engineering** (& Security) is an **investment**
  - **It has a cost** (no free lunch)
  - **It requires expertise**. And can build you a **career**.

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ENGINEERING

Computing & Software

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ITB 131 (appointment only)